
KUSHITA MADDELA

kushita07@gmail.com | <https://www.linkedin.com/in/kushitamaddela>

(313)-889-2941

Senior Software Engineer with 10+ years of experience in end-to-end software design, development, and system architecture. Expert in C++, C#, Python, ASP.NET, multithreaded real-time applications, device-level communication, SQL-based systems, and enterprise-grade distributed software. Proven track record of leading cross-functional teams and delivering production-quality software interfacing with electrical / embedded hardware, diagnostic systems, and cloud-connected device platforms. Strong background in TDD, OOP, SOLID, Agile, and CI/CD.

Professional Summary

- Proficient in C#, C++ and Python development on Linux platforms with strong debugging, performance optimization, and system-level programming expertise.
- Experienced in designing and implementing scalable APIs and inter-process communication mechanisms for data exchange across distributed systems.
- Strong SQL experience including database schema design, stored procedures, indexing, query optimization, and performance tuning for large-scale, real-time systems in MySQL and Postgres.
- Hands-on experience with containerization and deployment using Docker and Kubernetes, improving application scalability and maintainability.
- Integrated AI-based diagnostic and anomaly detection models using Python and deployed inference modules in C++ for real-time analytics in embedded systems.
- Strong understanding of networking and communication protocols including Multicasting (IGMP), DHCPv4/v6 Relay, ICMPv6 Relay, Dot1x, and IPv4/v6 Security.
- Skilled in full-cycle software development from requirements analysis and architecture design to implementation, testing, and deployment following Agile methodologies.

Technical Skills

Languages: C++, C#, Python, C, SQL, JavaScript

Frameworks & Libraries: PyTorch, TensorFlow, ONNX Runtime, TensorRT, OpenCV, NumPy, Pandas

Networking Protocols: DHCPv4/v6 Relay, ICMPv6 Relay, Dot1x, IPv4/v6 Security, Multicasting (IGMP), UDS, ARP, VRRP **Platforms:** Windows, Linux, RTOS, Android, Mac

Platforms: Embedded Linux, Windows, Android, RTOS, Mac

Tools & Technologies: Git, Docker, Kubernetes, Wireshark, Vehicle Spy, CANoe, GDB, Valgrind, Cereal, ClearCase

APIs & Communication: RESTful APIs, IPC mechanisms, MQTT, socket programming

DevOps & Deployment: Containerization, microservices, cloud orchestration (Docker/Kubernetes)

Professional Experience

Software Engineer, Flexera Global

May 2022 – Present

(Client: General Motors)

- Led a team of 7 engineers in developing a high-performance enterprise diagnostics platform using C++ and Python, improving operational efficiency and reducing project turnaround time by 20% organization wide.
- Designed and implemented an advanced Over-the-Air (OTA) update system leveraging Unified Diagnostic Services (UDS) in C++, enabling automated ECU software updates and achieving \$500,000 in annual cost savings.
- Refactored diagnostic communication layers from HIDL to AIDL, simplifying inter-process communication, improving interface stability, and reducing reported defects by 20%.
- Implemented secure APIs and cryptographic signing mechanisms for OTA packages, ensuring end-to-end data integrity, authenticity, and compliance with automotive cybersecurity standards.
- Developed and optimized C#/.NET Web APIs for high-throughput diagnostic services, implementing secure request handling, async pipelines, and connection-pool tuning to improve response efficiency and reliability across distributed systems.
- Developed and debugged SQL workflows across development, staging, and production environments, ensuring data accuracy, consistency, and performance throughout the release lifecycle.
- Integrated AI-driven analytics pipelines in Python to process ECU telemetry and detect anomalies, supporting predictive diagnostics and early fault identification.
- Containerized C++ and Python microservices using Docker and orchestrated deployments on Kubernetes, improving scalability, maintainability, and consistency across development and production environments.

Technical Lead, Nokia

Mar 2018 – Apr 2022

- Developed multithreaded C++ applications for configuration management and packet processing, improving system responsiveness and throughput.
- Designed and implemented IPC communication in Python to synchronize configuration and packet processing tasks, improving coordination and performance across processes.
- Developed C++ modules to maintain packet counters and IP address data in the application database, ensuring accurate updates and efficient retrieval.

- Refactored C++ and Python modules to improve performance by 30%, reducing execution time and enhancing overall system responsiveness.
- Migrated key application components from C++ to Python, reducing development time by 25% and simplifying future enhancements.
- Implemented SQL procedures and automation to purge dynamically learned IP entries on interface state transitions, guaranteeing accurate runtime routing and configuration persistence.
- Utilized Cereal libraries to serialize and deserialize application data, improving device availability and accelerating packet processing during system reboot.
- Developed ICMPv6 application to install the protocol filter for validating the ICMPv6 packet types in the lower layers.

Senior Software Engineer, Aricent

Dec 2014 – Mar 2018

- Developed C++ Layer-3 network protocol modules compliant with RFC standards, ensuring reliable and standards-based communication.
- Designed and tuned SQL procedures, indexes, and data validation logic to support protocol workflows, reducing configuration errors and improving system stability by 20%.
- Collaborated with QA and customer support teams to reproduce and analyze field issues, delivering quick turnaround fixes for live network environments.
- Gained hands-on exposure to high-availability architectures and In-Service Software Upgrade (ISSU) mechanisms to ensure uninterrupted network operation.
- Used Python to interact with internal and external APIs for network configuration, status monitoring, and diagnostics, streamlining system communication.